Guided Problem Solving - 5

Foundations of Fluid Mechanics, AM2530

September 22, 2026

1. The velocity field $\vec{V} = Ax\,\hat{i} - Ay\,\hat{j}$ represents flow in a "corner", where $A = 0.3\ \text{s}^{-1}$ and the coordinates are measured in meters. A square is marked in the fluid as shown at $t = 0$. Evaluate the new positions of the four corner points when point a has moved to $x = \frac{3}{2}$ m after τ seconds. Evaluate the rates of linear deformation in the x and y directions. Compare the area $a'b'c'd'$ at $t = \tau$ with the initial area $abcd$ at $t = 0$, and comment on the significance of this result.

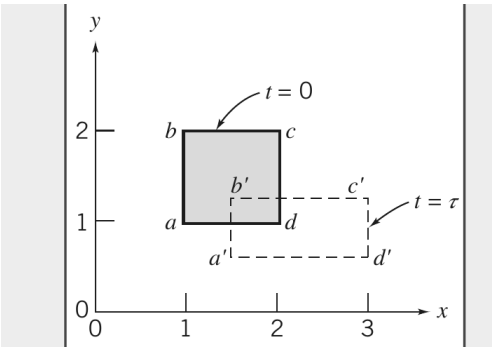


Figure 1

2. A viscous liquid is sheared between two parallel plates separated by $h = 4$ mm, where the lower plate is fixed and the upper plate moves with velocity $U = 4$ mm/s. Two perpendicular line segments ac and bd are marked in the fluid at $t = 0$ as shown. Plot the positions of these lines at $t = 1.5$ s. Determine the rate of angular deformation and the rate of rotation ω_z of the fluid element. If a small paddle wheel is pinned at its center in this flow, determine whether it will rotate, and explain why rotation occurs even though the streamlines are perfectly straight.

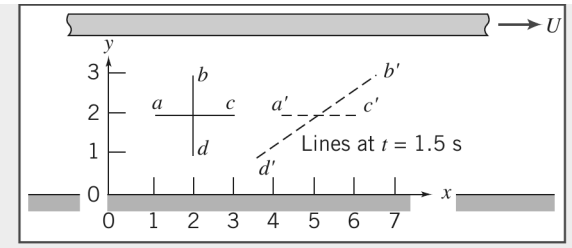


Figure 2

3. Calculate the mean velocities for these two-dimensional velocity profile geometries across a channel of height $2h$ if the centerline velocity is $v_c = 3$ m/s. Evaluate the rates of shearing strain $\frac{du}{dy}$ and the vorticity ζ_z across the channel for profiles (a) and (d), and identify the locations where each flow experiences maximum deformation.

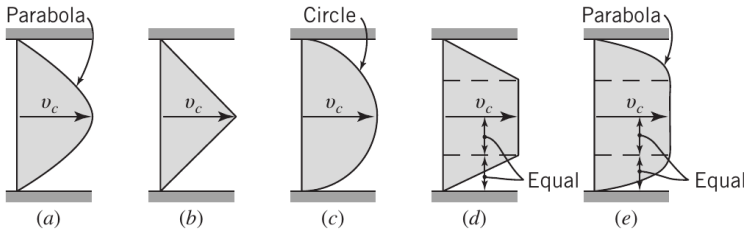


Figure 3

4. The flow between stationary parallel plates separated by distance $2b$ is driven by a pressure gradient. The velocity field is given by $u = U\left[1 - \left(\frac{y}{b}\right)^2\right]$, where U is the centerline velocity and y is measured from the centerline. Evaluate the rates of linear and angular deformation. Obtain an expression for the vorticity vector $\vec{\zeta}$, and find the location where the vorticity is a maximum. Show that the wall shear stress is directly proportional to the wall vorticity, $\tau_w = -\mu\,\zeta_z(b)$.

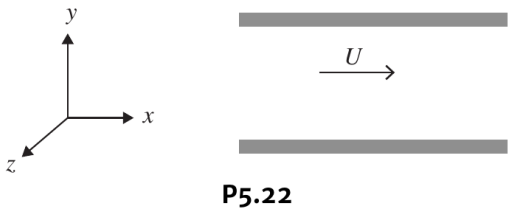


Figure 4

5. There is a fully developed laminar flow of an incompressible fluid between two infinite parallel plates separated by a distance $2h$ as shown in the figure. The top plate moves with a velocity V_0 and there is a pressure gradient $\frac{dp}{dx}$. The velocity profile is $u(y) = \frac{V_0}{2}\left(1 + \frac{y}{h}\right) + \frac{1}{2\mu}\left(-\frac{dp}{dx}\right)(h^2 - y^2)$. Determine the pressure gradient and the rate of rotation Ω_z for which the net volume flow rate is zero. Identify the location in the channel where the rate of deformation is zero.



Figure 5

6. Water flows steadily at 3 m^3/s through a 1 m diameter pipe. The stream is deflected outwards and downwards at 45° through an annular opening by a curved mushroom deflector as shown. Find the exit discharge speed V. Using the differential continuity equation in cylindrical coordinates, show that the vertical velocity gradient $\frac{\partial V_z}{\partial z}$ vanishes in the radial turning stream, and explain how the spreading annular sheet preserves in-plane surface area.

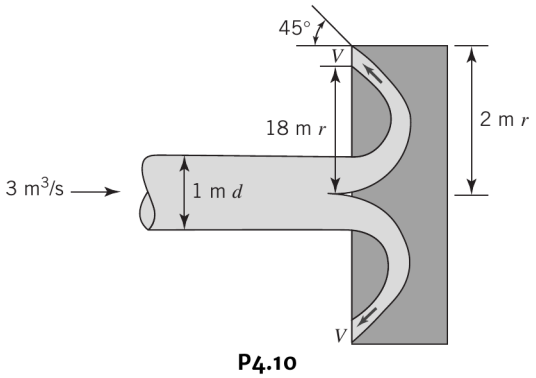


Figure 6

7. For the laminar boundary layer flow of an incompressible fluid over a flat plate ($y = 0$), the streamwise velocity is approximated by:

$$u(x, y) = U \frac{y}{\delta(x)} \quad \text{for } 0 \leq y \leq \delta(x),$$

where U is the freestream velocity and the boundary layer thickness grows downstream as $\delta(x) = c\sqrt{x}$ (c is a constant), as depicted in Figure 7. Assuming flow to be incompressible, derive an analytical expression for the transverse velocity component $v(x, y)$. At a section where $x = 0.5$ m and $\delta = 5.0$ mm, evaluate the velocity ratio v/U at the wall ($y = 0$), at the mid-layer ($y = \delta/2$), and at the outer edge ($y = \delta$). Explain physically why incompressibility requires a non-zero velocity normal to the plate.

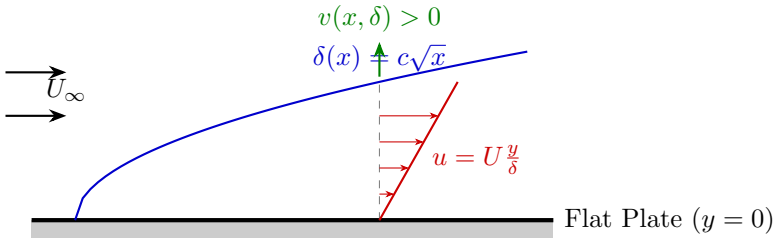


Figure 7: Laminar boundary layer growth and transverse velocity displacement due to incompressibility.